



DATA SCIENCE FIELD NOTES / 03B

STATISTICAL ANALYSIS REPORT

Six hypothesis tests separating strong evidence, weak signals and unsupported differences

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PROJECT	Loan Approval Analysis and Feature Engineering
PHASE	Week 3 Completed analytical preparation

Educational portfolio report. Historical approval patterns do not constitute a production lending model or lending policy.

Analytical scope

Six hypothesis tests evaluate categorical associations, differences between approved and rejected groups, geographic income variation and the relationship between household income and requested loan amount.

Analytical element	Application
Significance threshold	$\alpha = 0.05$
Categorical association	Chi-square test of independence
Two skewed numerical groups	Mann-Whitney U test
Three skewed numerical groups	Kruskal-Wallis test
Monotonic numerical relationship	Spearman rank correlation
Categorical effect size	Cramer's V

Assumption assessment

Income and loan variables are strongly right-skewed and contain plausible extreme observations. Normality checks and visual inspection support nonparametric rank-based comparisons. Expected contingency-table counts exceed common minimum thresholds for the chi-square tests.

Interpretation rule: a small p-value supports an association or group difference. It does not establish causation, business importance, fairness or legal permissibility.

Test 1 - Credit history and approval

H0: credit history and approval are independent. H1: they are associated.

Credit history	Rejected	Approved	Approval rate
0 - No positive history	82	7	7.87%
1 - Positive history	110	415	79.05%

Result: chi-square = 176.115; df = 1; p = 3.418e-40; Cramer's V = 0.536; minimum expected count = 27.83. Reject H0. Credit history has a strong observed relationship with historical approval.

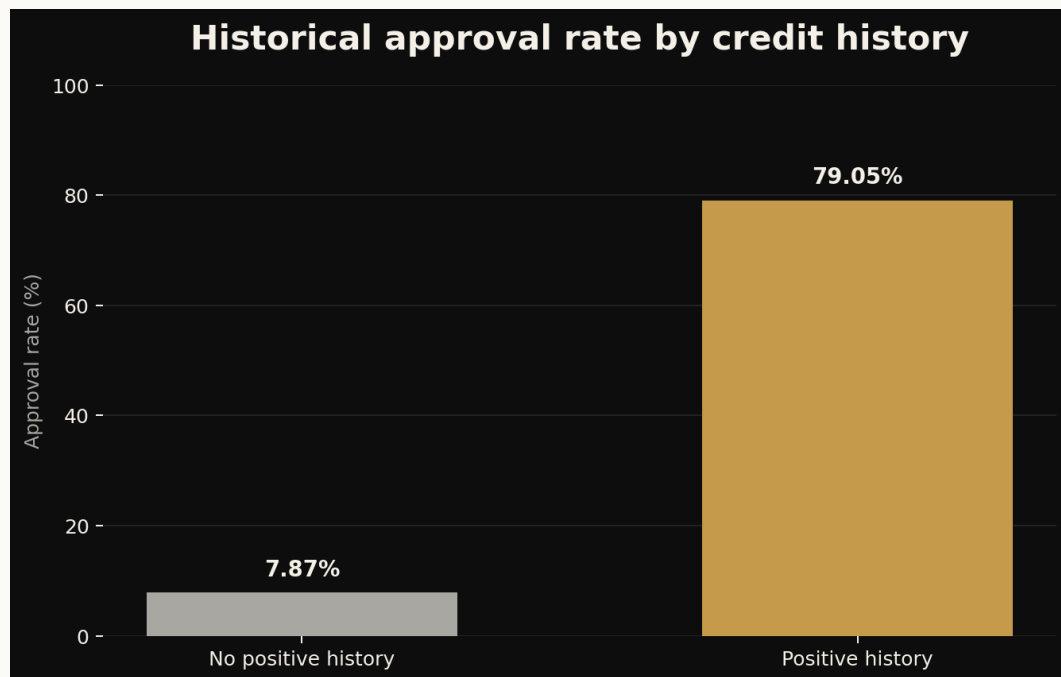


Figure 1. The largest observed approval-rate separation in the analysis.

Test 2 - Property area and approval

H0: property area and approval are independent. H1: they are associated.

Property area	Rejected	Approved	Approval rate
Semiurban	54	179	76.82%
Urban	69	133	65.84%
Rural	69	110	61.45%

Result: chi-square = 12.298; df = 2; p = 0.0021; Cramer's V = 0.142. Reject H0, but treat property area as a weak secondary signal.

Test 3 - Total income by approval outcome

H0: total-income distributions are equivalent across outcomes. H1: the distributions differ.

Outcome	Observations	Median total income
Approved	422	5,439.00
Rejected	192	5,289.50

Mann-Whitney U = 41,262.5; $p = 0.7128$; rank-biserial effect = -0.019. Do not reject H0. Total income does not independently distinguish historical outcomes in this sample.

Test 4 - Loan amount by approval outcome

H0: requested loan-amount distributions are equivalent across outcomes. H1: the distributions differ.

Outcome	Observations	Median loan amount
Approved	422	128.00
Rejected	192	128.00

Mann-Whitney U = 38,788.0; $p = 0.3976$; rank-biserial effect = 0.043. Do not reject H0. Loan size should not be interpreted in isolation.

Non-significance does not prove that a feature is useless. Income and amount may contribute through interactions or engineered relationships.

Test 5 - Total income across property areas

H0: rural, urban and semiurban applicants have equivalent total-income distributions. H1: at least one group differs.

Property area	Observations	Median total income
Rural	179	5,704.00
Semiurban	233	5,191.00
Urban	202	5,258.50

Kruskal-Wallis H = 3.933; p = 0.1400. Do not reject H0. The observed area-level approval differences cannot be attributed solely to distinguishable total-income distributions.

Test 6 - Income and requested loan amount

H0: there is no monotonic relationship between total household income and requested loan amount. H1: a relationship exists.

Spearman rho = 0.688; p = 3.423e-87. Reject H0. Higher-income applicants generally request larger loans. The result supports income-relative affordability features but does not show that income causes approval.

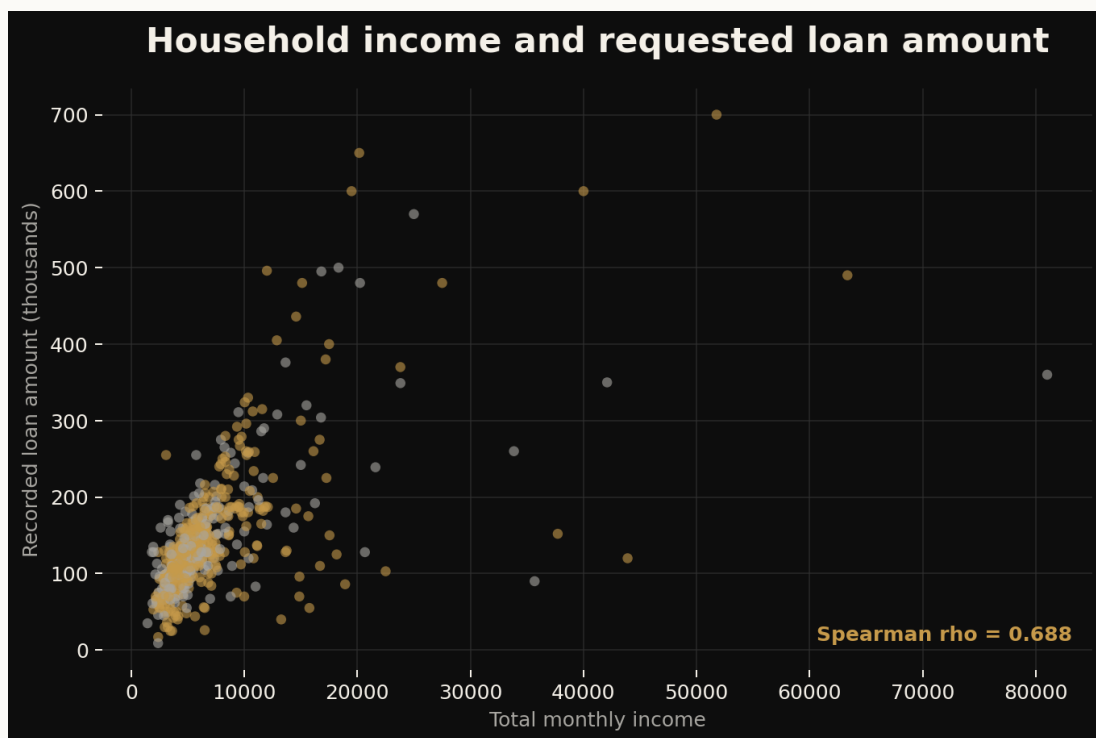


Figure 2. Strong positive monotonic relationship between household income and requested amount.

Consolidated statistical results

Analysis	Statistic	p-value	Decision
Credit history vs approval	176.115	<0.001	Reject H0
Property area vs approval	12.298	0.002	Reject H0
Total income by approval	41,262.5	0.713	Do not reject
Loan amount by approval	38,788.0	0.398	Do not reject
Income across property areas	3.933	0.140	Do not reject
Income vs loan amount	0.688	<0.001	Reject H0

Effect size and practical significance

Credit history combines a very small p-value with a comparatively strong effect size. Property area is statistically significant but much weaker, so it should not receive equal operational weight. The income and loan-amount group comparisons have very small rank-biserial effects.

Interpretation controls

- Multiple tests were conducted; borderline results require cautious interpretation and later validation.
- The target records historical approvals rather than repayment or default.
- Non-significance does not rule out predictive value in interaction with other variables.
- Statistical association does not establish a causal or legally permissible lending criterion.

Conclusion

The evidence supports prioritising credit history, retaining property area as a secondary candidate, engineering income-relative affordability indicators and validating every conclusion through out-of-sample modelling and fairness checks.